Quantitative Data Analyst Intern - Take-Home Assessment

Objective

Analyze how corporate announcements affect stock price and trading volume. The exercise evaluates financial-data cleaning, event alignment, statistical reasoning, Python proficiency, and the ability to communicate useful findings.

You are not expected to build a trading system or a complex machine-learning model. A simple, correct, and well-explained analysis is preferred over a large or overly complicated solution.

Input

We will provide:

- `corporate_announcements.csv`, containing BSE corporate announcements for five companies.
- `RELIANCE.csv`, `HDFCBANK.csv`, `NYKAA.csv`, `HAL.csv`, and `RVNL.csv`, containing one-minute OHLCV data for the corresponding stocks.
- `DATA_FORMAT.md`, containing the field definitions, symbol mapping, coverage, timestamp convention, and data caveats.

The announcement dataset contains fields such as the announcement ID, company, subject, category, subcategory, text, and dissemination timestamp. The market files contain `date`, `open`, `high`, `low`, `close`, `volume`, and `oi`.

Assignment

1. Prepare and Validate the Data

- Parse and align announcement and market timestamps correctly. Use `DissemDT` when it is present and valid, with `DT_TM` as the documented fallback.
- Map each announcement to the first complete one-minute bar beginning at or after dissemination. Avoid using a bar that partly occurred before the news.
- Correctly handle announcements made before market open, during the session, after market close, on weekends, and on non-standard trading sessions.
- Check for missing or duplicate records, invalid OHLC values, zero-volume bars, incomplete event windows, and other relevant data-quality issues.
- Record material cleaning decisions and exclusions. Do not silently discard inconvenient observations.
- Detect announcements that relate to the same underlying event. For example, financial results, a board-meeting outcome, a press release, and an investor presentation published close together should not automatically be treated as four independent events.

You may choose your own event-clustering rule, but it must be explained and applied consistently.

2. Create an Announcement Subject Taxonomy

- Use the supplied category, subcategory, subject, headline, and narrative fields to create a practical set of announcement subjects.
- Merge inconsistent labels where appropriate and distinguish routine filings from potentially price-sensitive announcements.
- Analyze at least five meaningful subject groups.
- Report the number of independent events in every group. Treat groups with fewer than 10 independent events as descriptive rather than statistically conclusive.

A transparent rule-based classification is sufficient. Complex NLP is not required.

3. Measure Price and Volume Impact

For each usable event, measure the stock's response over at least these windows:

- First 5 minutes after the event becomes tradable.
- First 30 minutes.
- First 60 minutes.

- From the event to that trading session's close.
- 1, 5, 10, and 20 trading sessions after the event.

For announcements that become tradable at the next session's open, include the overnight gap in the event-day response.

Your subject-wise analysis must include:

- Return direction and magnitude.
- Change in volume relative to a suitable historical baseline.
- At least one measure of volatility or price range.
- Mean, median, sample size, and a measure of uncertainty or statistical significance.
- A comparison of impact across subjects and stocks.
- A clear definition of what you consider a strong announcement impact.

Do not assign numeric codes to subject labels and calculate an ordinary correlation from those codes. Measure the relationship between announcement subject and the resulting price-volume response using a suitable comparison, test, or model.

4. Mandatory Post-Earnings Announcement Drift Analysis

Identify independent financial-result announcement events and investigate whether their initial price response is followed by a continued move in the same direction.

For this assessment, use a price-action-based definition:

- Define `D0` as the first trading session in which the earnings information could be acted upon.
- For an in-session release, measure the initial reaction from the last pre-announcement price through the `D0` close. For a release that first becomes tradable at the open, measure from the previous regular-session close through the `D0` close.
- Use the sign of the `D0` return as the market-implied direction of the news.
- Measure subsequent returns through `D+1`, `D+3`, `D+5`, `D+10`, and `D+20`.
- Determine whether the subsequent move continues, reverses, or becomes indistinguishable from normal variation.

At minimum, report:

- The amount of post-announcement drift at each horizon.
- The percentage of events that continue in the initial direction.
- A comparison of the initial reaction with the later drift.
- The approximate duration for which any drift remains visible.
- Pooled results across all stocks and a stock-wise breakdown.
- Sample sizes and uncertainty around the estimates.

Under-reaction is a hypothesis to test, not an assumed result. A well-supported finding that drift is weak, inconsistent, or absent is a valid conclusion. Because analyst-expectation or earnings-surprise data is not supplied, describe this as price-conditioned post-earnings announcement drift rather than making a claim about surprise-based PEAD.

5. Explain the Findings

Present 5-10 concise insights. Each important insight should state:

- What you observed.
- The size and direction of the effect.
- The sample size or statistical support.
- Why the finding matters.
- The main limitation or alternative explanation.

Minimum Statistical Expectations

- Report both mean and median where outliers can affect the result.
- Include sample sizes with all grouped results.
- Use at least one appropriate confidence interval, hypothesis test, or resampling method for the main subject-impact and drift conclusions.
- State how overlapping events and extreme observations were handled.

- Do not claim causality or generalize beyond the supplied companies without appropriate qualification.

Advanced econometrics, machine learning, sentiment models, and strategy backtests are optional and should be attempted only after the required analysis is complete.

Required Deliverables

Publish the completed solution in a public GitHub repository containing:

1. `README.md`
   - Brief methodology and important assumptions.
   - Setup instructions and a single clear way to reproduce the analysis.
   - A short list of the main findings and limitations.
2. Python source code and/or notebooks
   - The complete analysis must run from the original provided files.
   - Repeated manual spreadsheet steps are not considered reproducible.
3. Environment file
   - A candidate-created `requirements.txt`, `pyproject.toml`, or equivalent.
4. Short report
   - Submit `report.md`, `report.html`, or `report.pdf`.
   - Keep the report focused; approximately 8 pages or 2,500 words is sufficient.
5. Figures
   - Export 5-8 important charts as PNG or SVG files under `outputs/figures/` and include clear titles, axes, units, and captions.
6. Result tables
   - `outputs/event_level_results.csv`: event timestamp, effective event time, subject, event cluster, key return-volume metrics, and exclusion reason.
   - `outputs/subject_summary.csv`: subject-wise impact statistics.
   - `outputs/pead_summary.csv`: earnings-drift results by horizon and stock.

Important

- Python is mandatory. You may use pandas or Polars, NumPy, SciPy, scikit-learn, statsmodels, and visualization libraries of your choice.
- Do not commit the supplied datasets to the public repository. Add the provided data paths to `.gitignore` and document where the files should be placed to reproduce the analysis.
- External data is not required. If you use any, cite its source and explain why it was needed.
- Use Git throughout the exercise and maintain a clear, meaningful commit history.
- State any use of AI-assisted tools and be prepared to explain all submitted code and analysis.
- We value correct timestamps, defensible assumptions, reproducibility, and honest interpretation more than the number or complexity of techniques used.

Evaluation Criteria

Area	Weight
Data validation and event alignment	15%
Event clustering and subject taxonomy	10%
Price-volume impact analysis	20%
Post-earnings announcement drift analysis	20%
Statistical reasoning	15%
Code quality and reproducibility	10%
Insight quality and communication	10%